

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/341891321>

Total electron content anomalies associated with earthquakes occurred during 1998–2019

Article in *Acta Astronautica* · June 2020

DOI: 10.1016/j.actaastro.2020.06.005

CITATIONS

15

READS

240

7 authors, including:



Munawar Shah

Institute of Space Technology

51 PUBLICATIONS 307 CITATIONS

SEE PROFILE



Arslan Ahmed

Newcastle University

23 PUBLICATIONS 146 CITATIONS

SEE PROFILE



Dr Muhsan Ehsan

Bahria University

23 PUBLICATIONS 121 CITATIONS

SEE PROFILE



Majid Khan

University of Science and Technology Beijing

34 PUBLICATIONS 169 CITATIONS

SEE PROFILE

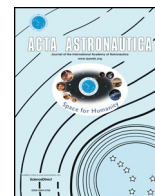
Some of the authors of this publication are also working on these related projects:



Geophysical Research on Rockburst Identification and Prediction through Mapping Induced Fractures in Deep Coal Mines [View project](#)



Geodesy Teaching Materials [View project](#)



Total electron content anomalies associated with earthquakes occurred during 1998–2019



Munawar Shah^{a,h,*}, Arslan Ahmed^b, Muhsan Ehsan^c, Majid Khan^d, Muhammad Arslan Tariq^e, Andres Calabia^f, Zahid ur Rahman^g

^a Department of GNSS, Institute of Space Technology, Islamabad, Pakistan

^b Department of Electrical Engineering, Sukkur Institute of Business Administration, Pakistan

^c Department of Earth and Environmental Sciences, Bahria University Islamabad, Pakistan

^d School of Civil and Resource Engineering, University of Science and Technology Beijing, 100083, Beijing, China

^e Centre for Earthquake Studies, National Centre for Physics, Pakistan

^f School of Remote Sensing and Geomatics Engineering, Nanjing University of Information Science and Technology, Nanjing, 210044, China

^g Department of Computer Science, Bacha Khan University, Charsadda, KPK, Pakistan

^h Space Education and GNSS Lab, National Center of GIS and Space Application, Institute of Space Technology, Islamabad, Pakistan

ARTICLE INFO

Keywords:

Lithosphere- ionosphere mechanism

Earthquakes

GNSS TEC

Geomagnetic storm

ABSTRACT

Statistical analyses on the intensification of seismic precursors before and after the earthquakes (EQs) from Global Navigation Satellite System (GNSS) based Total Electron Content (TEC) may provide evidences about the potential mechanisms of lithosphere-ionosphere coupling. In this paper, ionospheric anomalies before and after 1182 ($M_w > 5.0$) EQs during 1998–2019 at three different latitudinal zones (80°N–25°N, 25°N–25°S, and 25°S–80°S) are investigated from global GNSS TEC observations. Previous studies proposed different time intervals for the occurrence of TEC anomalies triggered by EQs. In this study, we reveal that prominent TEC anomalies in the form of enhancement and depletion occur within 5 days before and after the main shock of the EQs. Further analysis suggests that the probability of TEC anomalies occurred before EQs of $M_w \geq 6.0$ and focal depth of less than 220 km (Zone B) is about 0.8. Similarly, a probability of 0.65 for TEC anomalies is recorded before EQs of $M_w \geq 7.0$ and focal depth less than 120 km in Zones-A and C. In our analysis, TEC anomalies generated by EQs are isolated from those produced by the geomagnetic storms through the K_p geomagnetic activity index and the disturbance storm time (Dst) index.

1. Introduction

The existence of lithosphere-ionosphere coupling before and after strong EQs is an intriguing phenomena still not understood which has been widely reported in the literature [1–3]. Among these studies, anomalous EQ perturbation from statistical analysis have shown clear insights on the generation of ionospheric perturbations before and after strong EQs [4–6]. Moreover, ionospheric anomalies are discussed in details from the analysis of ground and space measurements [7]. Liu et al. [8] first implemented a statistical approach with TEC estimations over the epicenter of the 1999 Chi-Chi event, and found significant perturbations in the ionosphere. Later on, a number of statistical studies have reported abnormal behavior of ionospheric TEC subject to EQs [9,10], and the modeling of ionospheric data has been tested for the prediction of EQs [11]. In all these studies, the statistical analyses have provided a background to study the occurrence of ionospheric

anomalies as implemented by Liu et al. [12]. Similarly, ionospheric anomalies associated with large EQs in the form of statistical analysis have been reported from the Detection of Electromagnetic Emissions Transmitted from Earthquake Regions (DEMETER) satellite measurements in Shah et al. [13].

With the development of Global Navigation Satellite System (GNSS) technologies and its application in atmospheric sciences, specifically TEC retrieval, more and more studies have reported the enhancement of TEC before and after the main shock. Similarly, other reports have shown the anomalous enhancement in TEC during the EQ preparation period [14]. Moreover, Liu et al. [12] observed the anomalous increase in TEC in a statistical analysis performed for EQs with $M_w \geq 5.0$ during 1994–1999 in Taiwan, and determined that TEC perturbations could only be seen for large magnitude and shallow depths EQs. They further emphasized on the importance of statistical analysis for possible links of TEC anomalies over the EQ breeding zone. Then, many researchers

* Corresponding author. Department of GNSS, Institute of Space Technology, Islamabad, Pakistan.

E-mail addresses: Munawar.shah@mail.ist.edu.pk, shahmunawar1@gmail.com (M. Shah).

<https://doi.org/10.1016/j.actaastro.2020.06.005>

Received 25 January 2020; Received in revised form 4 May 2020; Accepted 2 June 2020

Available online 03 June 2020

0094-5765/ © 2020 IAA. Published by Elsevier Ltd. All rights reserved.

have performed rigorous statistical analysis using ground and satellite-based ionospheric estimations [1,15–18].

Many other reports have shown different aspects of the ionospheric anomalies induced by EQs on the global and local scales. For example, Astafyeva et al. [19] reported that the enhancement in TEC before large magnitude EQs occurred along strike-slip faults. Similarly, Shah and Jin [3] investigated the TEC anomalies associated with different fault ruptures during 1998–2014, and suggested that strike-slip fault related anomalies have significant peak within five days before the main shock as compared to other fault system EQs. Furthermore, Zhu et al. [6] showed enhancement in the nighttime TEC from Global Ionospheric Maps (GIMs) as compared to daytime for 1339 globally $M_w > 6.0$ EQs during 2003–2014. They also found a clear enhancement in the occurrence rate of positive and negative TEC anomalies with increasing EQ magnitude during the day and nighttime observations. And similarly, more reports also showed statistical analysis of seismo ionospheric anomalies from GIM TEC [12,14,17]. On the other hand, some other studies have shown the lack of significant relationship between EQs and ionospheric anomalies [20,22], and even correlated the ionospheric anomalies with geomagnetic storms rather than EQs [21].

In this paper, a detailed statistical analysis of GNSS TEC anomalies on the basis of spatial and temporal deviations is performed over the epicenter associated with EQs of 1182 ($M_w > 5.0$) EQs during 1998–2019. Quiet geomagnetic conditions are considered for all the events. Similarly, TEC anomalies related to any other EQ within ten days before and after the studied event are also omitted. Based on the different morphological characteristics of the EQs, the anomalous values of TEC at different latitudinal regions are investigated. The main aim of this study is to help in future for monitoring of ionospheric anomalies around the seismically active regions on Earth for successful EQ forecasting system. This manuscript is structured as follows: next section introduces the data and methods employed in this study; section 3 provides description of results and discussion in reference to previous authors; finally, conclusions is given in the last sections.

2. Data and method

The statistical analysis of seismo ionospheric anomalies by the enhancement and depletion of TEC related to 1182 EQs of $M_w > 5.0$ during 1998–2019 is divided in three different latitudinal zones (80°N – 25°N , 25°N – 25°S , and 25°S – 80°S). In this study, the ionospheric perturbations and its different morphological characteristics related to EQs are investigated before and after the seismic events. The different morphological parameters of the EQs include magnitude, depth, and location, and the data is retrieved from the website of the United States Geological Survey (USGS) <http://earthquake.usgs.gov/earthquakes/search/>. The 1182 EQs included in this study are declustered from aftershocks following the method of Michael [23] and Thomas et al. [5]. The resulting geographical location of the declustered epicenters is shown in Fig. 1.

Along with different EQ parameters, we have obtained space weather conditions for each event. We consider the effects of geomagnetic storms in TEC variability and employ the geomagnetic storm activity index (K_p) and the disturbance storm time index (Dst) available at the OMNI web NASA and the Kyoto University websites (<http://wdc.kugi.kyoto-u.ac.jp/dstdir/index.html>). Fig. 2 shows an example of the magnetospheric conditions before and after the February 21, 2017, Bolivia EQ. Therefore, the 1182 EQs employed in this study have been screened depending on the geomagnetic storm conditions.

We employ hourly TEC combined product from the International GNSS Service (IGS) [24–26], to extract the vertical TEC (VTEC) at the location of each EQ epicenter. Each VTEC map comprises a longitudinal extension of $\pm 180^\circ$ and a latitudinal extension of $\pm 87.5^\circ$ [37]. The grids are provided at $5.0^\circ \times 2.5^\circ$ resolution in longitude and latitude, respectively [38].

Several studies have shown seismo ionospheric anomalies using

different methods of detection, including quartile, median, moving averages and sliding window, and principal component analysis techniques [3,12,16,27,34]. Since, there is no strict criteria for the ionospheric anomaly detection and no method provides an explicit definition of anomalies. In this study, after TEC values are obtained at each EQ epicenter, the variation in TEC is investigated through the confidence bounds of median (μ) and associated inter-quartile range (IQR). This statistical procedure was employed in Liu et al. [12] to identify anomalous signals in TEC. We compute the bounds from the median (μ) and IQR of 15 days before and after the observed day. Then calculate the deviation between the observed value for each EQ diurnal TEC and its confidence bounds. In the literature, several methodologies exist to calculate the confidence bounds based on the median and the standard deviation of TEC [3,16,28]. In this work, the confidence bounds are calculated as $\mu \pm 2 \times \text{IQR}$. We define the positive anomaly when the diurnal TEC occurs beyond the upper bound for a period longer > 3 h. Similarly, significant negative anomalies are defined for TEC below lower bound for more than 3 consecutive hours. TEC anomalies during suspected hours are investigated in Dobrovolsky et al. [28] region to correlate the lithosphere and ionosphere coupling around the seismogenic zone. The seismogenic zone is calculated as:

$$R = 10^{0.43M} \quad (1)$$

In this equation, M is the EQ magnitude and R is the radius (km) of affected zone, which increases exponentially with the EQ magnitude.

Since geomagnetic storms can also cause TEC anomalies, several studies have associated TEC variations to geomagnetic storms instead of EQs [22,29–31]. These reports stressed on the inheritance of TEC anomalies with geomagnetic storms, and specifically declared that geomagnetic storms induced reasonable TEC variations as compared to EQs. Therefore, it is necessary to remove from the statistical analysis those days associated with geomagnetic storms, so we exclude the days with $Dst < -50$ nT and $K_p > 3$. Geomagnetic storms usually have three phases, namely initial, main and recovery phases [32,33]. Then, we exclude the days which comprise the initial to the recovery phase of the geomagnetic storm. Similarly, abnormal TEC days beyond 10 days before and 10 after the EQs or geomagnetic storms are not counted in the statistical analysis. This scheme may provide a clear picture of the TEC variation triggered by either EQs or geomagnetic storm within 10 days before/after. As this study mainly focus on possible anomalies within the time interval of 10 days before and after the main shock.

3. Results and discussions

In order to quantify the seismo ionospheric anomalies, we first investigate as a case study the EQ on February 21, 2017 (M_w 6.5, Padilla Bolivia). The geographical location of its epicenter is 19.281°S and 63.905°W . This EQ hits the region at 17:09:04 (UTC) at a hypocentral depth of 50 km. The epicenter was located 41 km East of Padilla city, Bolivia, and occurred in the Zone-B of the study area. Fig. 3 shows the diurnal TEC for 31 days before and after the main shock, with the corresponding confidence bounds. A significant positive anomaly is seen over the epicenter on the day of the main shock. It can be noted here that we only count TEC anomalies within 10 days before and after the main shock of this EQ. Similar analysis is implemented for other 1182 EQs of $M_w > 5.0$. To provide a stringent proof of temporal TEC variations triggered by the M_w 6.5 Bolivia EQ, spatial TEC deviation is plotted over the epicenter in Fig. 4. The deviation in spatial TEC is obtained by subtracting the monthly TEC values from the observed TEC at a specific UT hour. We can see that the temporal plot in Fig. 3 has a positive anomaly on the EQ day for more than 4 h. TEC values larger than 10 TECU are also noted in the spatial maps during UT = 10–15 h over the EQ epicenter (Fig. 4). The deviation in TEC values over the epicenter is greater than 30% during these hours, relatively larger than the rest of the analyzed period.

The cumulative number of potential seismo ionospheric anomalies

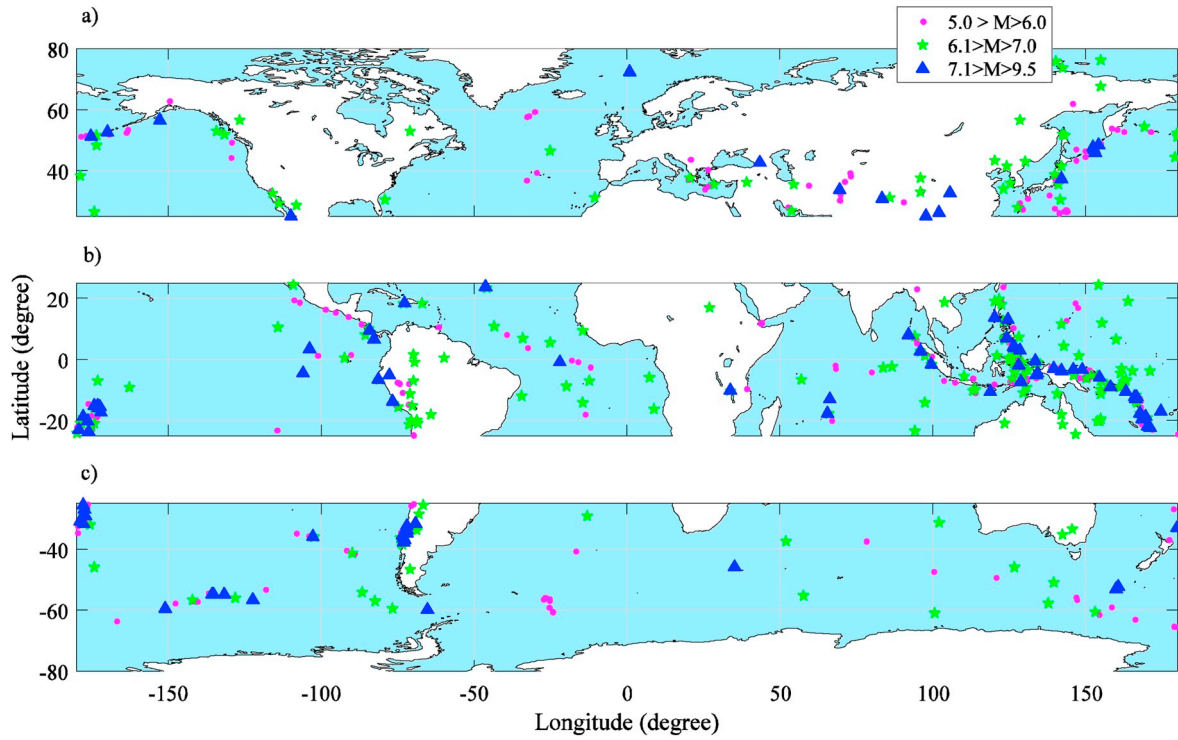


Fig. 1. Geographical distribution of the declustered EQs from aftershocks during 1998–2019 in 3 different latitudinal zones: in a) Zone-A, in b) Zone-B, and in c) Zone-C.

for the three latitudinal zones (A, B, and C) within 10 days before and after $M_w > 5.0$ EQs on Earth is plotted in Fig. 5. The cumulative number of TEC for a single EQ is obtained by dividing the TEC anomalies of a single day from the total TEC anomalies associated with all events. Similarly, the Index Number in percentage of the potential seismo ionospheric anomalies is obtained from the whole time-series by the following formula.

$$\text{Index No} = \frac{P_{SD}}{P_{Total}} \quad (2)$$

In this equation, P_{SD} and P_{Total} are the seismic precursors on a single day and total precursors associated with all EQs, respectively. Fig. 5 shows the cumulative number of ionospheric anomalies with abrupt enhancements within 5 days before the EQs of $M_w > 7.0$ in all the 3

zones. Moreover, in Zone-B, seismo ionospheric anomalies are associated with $M_w > 6.0$ during 5 days prior to the main shocks (Fig. 5; Panel 5f). Note that ionospheric anomalies due to geomagnetic storms are excluded in the analysis and storm induced anomalies are shown in the last Figure. The cumulative number of seismo ionospheric anomalies after removing the days of geomagnetic storm are obtained by the same method used in Shah and Jin [4].

The Index number in Zone-B for EQs of $6.1 < M_w < 7.0$ rises up to 8–10%, and about 15% for EQs of $M_w > 7.0$. Then, in general, the index number drops abruptly to 5% after 5 days of the EQs main shock. However, the analyses show that pre-seismic ionospheric anomalies within 5 days before the EQ event are more frequent than the post-seismic anomalies. Moreover, we can see that pre-seismic ionospheric anomalies become stronger before EQs of $M_w > 7.0$ in Zones-A and C.

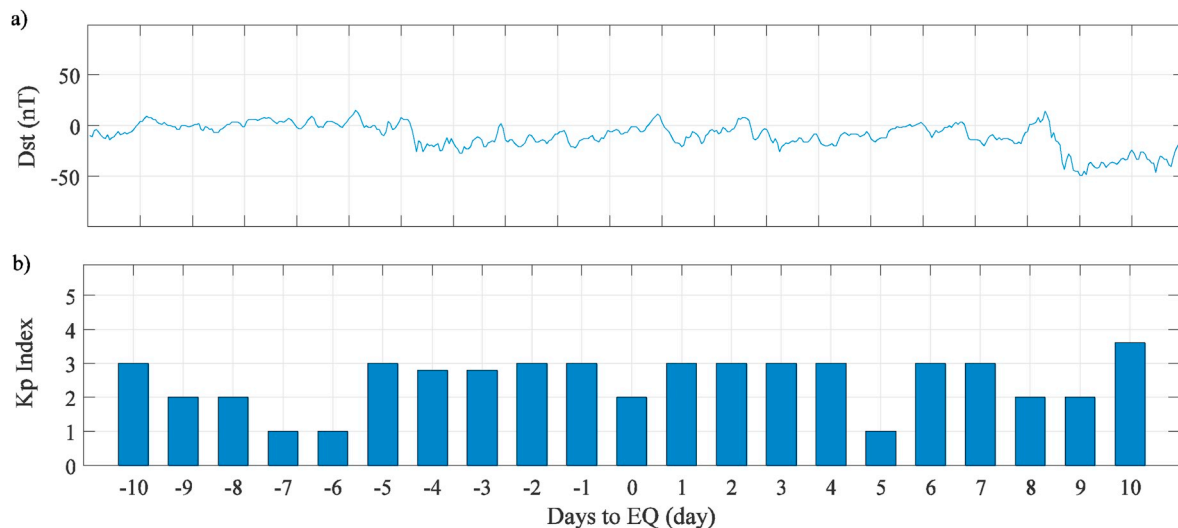


Fig. 2. Geomagnetic storm conditions before and after the February 21, 2017, M_w 6.5 Padilla, Bolivia EQ.

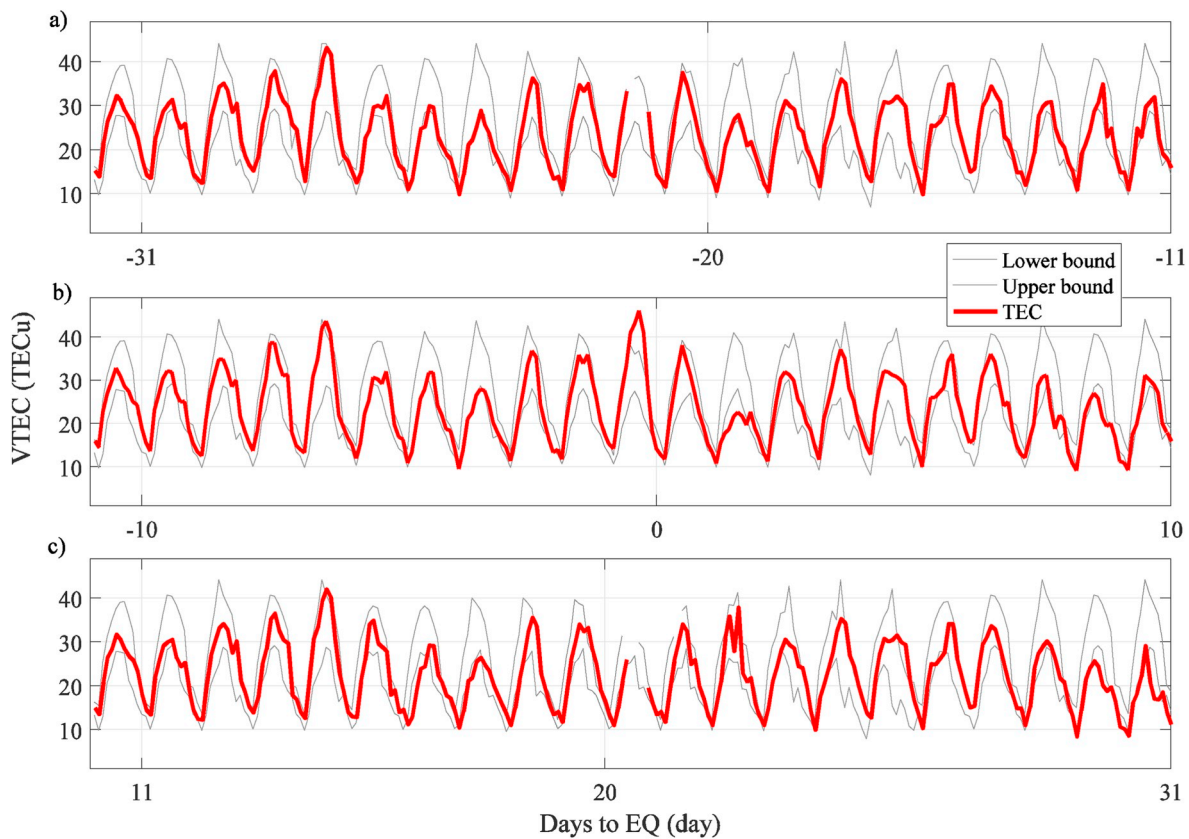


Fig. 3. The investigation of seismo ionospheric anomalies related to the February 21, 2017, M_w 6.5 Padilla, Bolivia EQ by temporal TEC within 31 days before and after the event. The diurnal TEC and confidence bounds are represented by a red line and two grey lines, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

The index values of pre-ionospheric anomalies for Zones-A and C before EQs of $M_w > 7.0$ are 7% and 10%, respectively (Fig. 5c and k). It seems that pre seismic TEC anomalies are more frequent in the Zone-B (Index $No \geq 10$ in Fig. 5e–h) than Zones-A and C. Similarly, it can be seen

that the number of seismo ionospheric anomalies is more frequent in Zone-B than Zones-A and C. Furthermore, the index number shows that EQs in Zone-B produce more ionospheric anomalies than the EQ events in the Zones-A and C. Ionospheric enhancements in equatorial regions

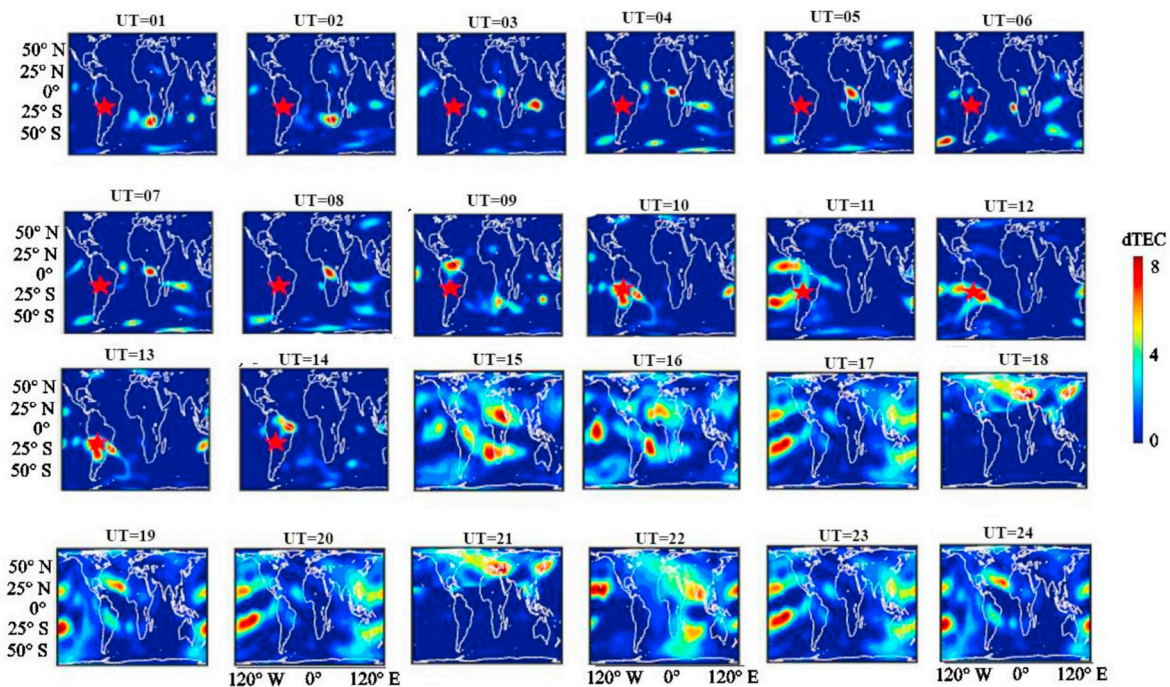


Fig. 4. The hourly spatial TEC maps over the epicenter of February 21, 2017, M_w 6.5 Padilla, Bolivia EQ showed positive anomalies during UT = 10 to UT = 15.

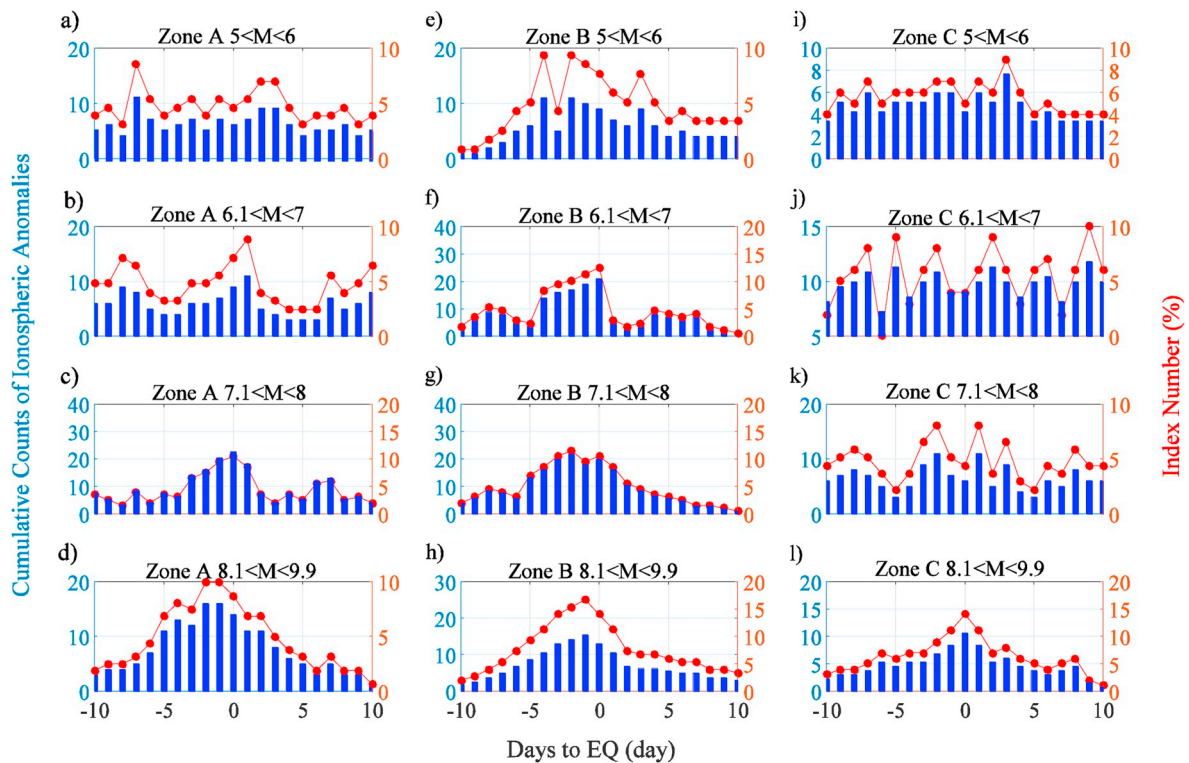


Fig. 5. The Cumulative counts and Index numbers of the seismo ionospheric anomalies during 10 days before and after the global EQs during 1998–2019.

(similar to Zone-B) are also reported in past in different case studies and statistical analysis [12].

Fig. 6 shows the enhancement of seismo ionospheric anomalies related to EQs of different focal depth in Zone-B as compared to Zones-A and C. Moreover, Fig. 6 also shows that TEC anomalies are associated with EQs having focal depth around 180–220 km in Zone-B, while anomalies occur with 100–120 km focal depth EQs in Zones-A and C. Fig. 6 also shows that seismo ionospheric anomalies are also dependent

on the hypocentral depth of EQs in Zones-A and C. Similarly, we can observe that the index number for the TEC anomalies in Zone-B is about 12% at 100 km depth, whereas, the index number at the same depth in Zones-A and C is around 8–10%. These TEC perturbations show a clear evidence of the precursory phenomena in ionosphere before EQs. Specifically, TEC anomalies in Zone-B are more frequent than the other Zones within 5 days before and after the main shock. Several studies have shown the existence of seismo ionospheric anomalies before and

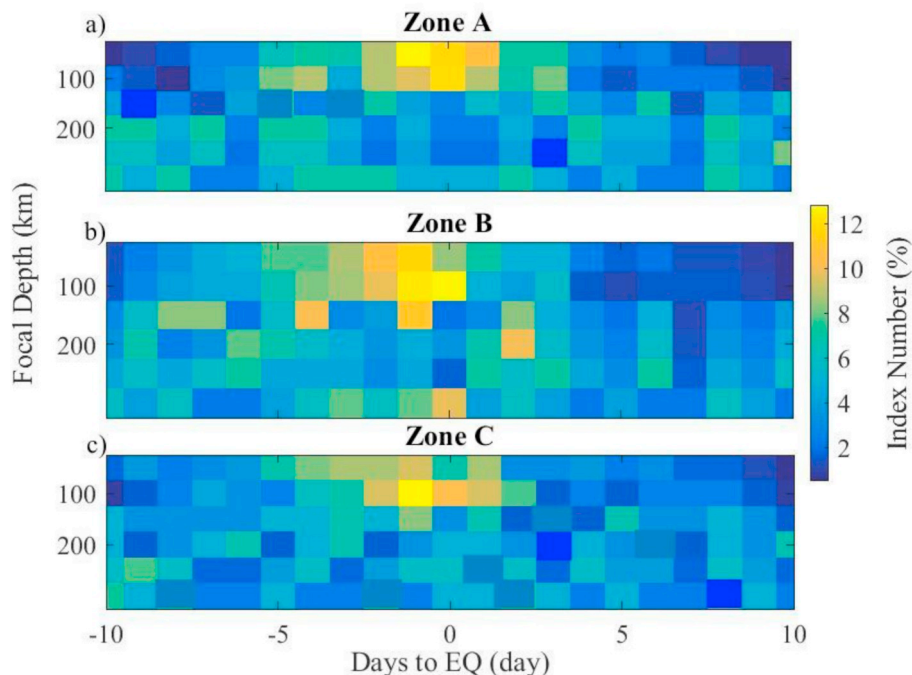


Fig. 6. The Index number analysis in percentages for the EQs in the three zones with different focal depths.

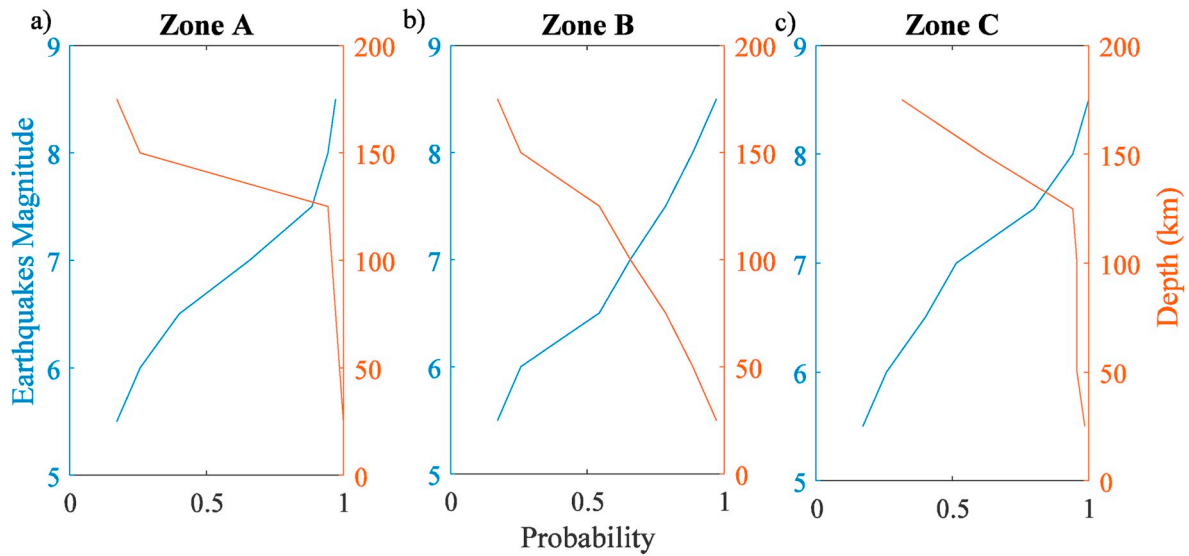


Fig. 7. The probability analysis for the threshold of TEC anomalies related to EQs in the three different zones of the world.

after EQs of shallow hypocentral depth [4,12,27]. Therefore, it is noteworthy to mention that in all these reports the existence of seismo ionospheric anomalies cannot be rejected, although none of them have presented an indisputable and clear definition of seismo ionospheric anomaly.

The study of seismo ionospheric anomalies for different zones shows the indisputable existence of TEC perturbation before the EQ events. On the basis of these results, we have computed the probability of TEC anomalies for the different EQ magnitudes and focal depths. The resulting threshold values are presented in Fig. 7. The probability is calculated by the number of TEC anomalies for a specific magnitude divided by the total number of TEC anomalies related to all events. Then, the probability is computed as follow:

$$\text{Probability} = \frac{\text{TEC}_{\text{Single EQ}}}{\sum \text{Total TEC Anomalies}} \quad (3)$$

We employ above formula to compute the probability for EQs of different magnitudes and focal depths which can generate seismic ionospheric anomalies. After implementing the probability statistics on TEC, the threshold magnitude for the generation of ionospheric anomalies in Zones-A and C is M_w 7.0. However, the threshold magnitude for Zone-B is M_w 6.0 (Fig. 7b). Similarly, the ionospheric abnormalities occurred with EQs of hypocentral depth 100–120 km in Zones-A and C (Fig. 7a & c) and the threshold depth for seismo ionospheric anomalies in Zone-B is 180–200 km (Fig. 7b). This equation and threshold is only valid for $M_w \leq 8.0$ EQs due to sample size problem, as EQs of M_w 5.0–8.0 induced more TEC anomalies. As EQs of large magnitude occurred less in number, Fig. 6 is only limited to M_w 8.5. The threshold probability can determine the magnitude and depth of the EQs which are likely to generate seismo ionospheric anomalies. Le et al. [14] showed the occurrence rate of seismo ionospheric anomalies for different EQ magnitudes and focal depths in a statistical analysis. In this paper, we have enlarged the data set for seismo ionospheric anomalies and investigated the probability to determine the threshold magnitude and focal depth during 1998–2019.

Furthermore, we study the deviation percentage of positive and negative anomalies from the mean distribution. We implement the analysis on the three zones for the association of EQ and ionospheric anomalies. The percentage deviation is computed as:

$$\% \text{ Dev} = \frac{\text{TEC}_{\text{Single EQ}} - \sum \text{Total TEC Anomalies}}{\sum \text{Total TEC Anomalies}} \times 100 \quad (4)$$

Fig. 8 shows more deviation from the mean distribution for positive

TEC anomalies than negative in all the 3 zones. This indicates that the enhancement of TEC related to EQs is more frequent than the depletion. The enhancement of TEC in the analysis of TEC over the EQ breeding zones was also reported by Shah and Jin [4]. Fig. 8b shows that the percentage deviation of positive and negative anomalies prior to the events in Zone-B during 5 days before the main shock is 80%, which is abnormal from the rest of the Zones. On the other hand, the anomalies in other zones are smaller than 50% within 5 days before and after the main shock. Note that the deviation from the mean is for positive and negative anomalies. Previous reports on seismo ionospheric anomalies also investigated positive and negative anomalies prior and after the main shock [4,6]. In Shah and Jin [4], the authors reported highest percentage of positive anomalies before $M_w > 5.0$ EQs. Similarly, this study also shows a high percentage of positive anomalies prior of $M_w > 6.0$ EQs in all the three zones.

Finally, the TEC response is examined during the geomagnetic storms of different strength during the same period (1998–2019). We examine TEC anomalies related to Dst and Kp indices. TEC anomalies are recorded before and after the different storms, and the abnormal days associated with these geomagnetic storm are excluded from the analysis of seismic ionospheric perturbations. Fig. 9 shows that the number of TEC anomalies due to high intensity geomagnetic storms is more than the number of anomalies associated with storms of low intensities. The reason is that the duration from the initial to final phase of high-intensity storm is longer than the low-intensity storms. The purpose of the study of TEC anomalies due geomagnetic storms is to determine the effect of space weather environment on the ionosphere. Furthermore it also assists in differentiating EQ and storm anomalies. There are several reports about the abnormal TEC values in the ionosphere related to geomagnetic storms [21,29,35]. However, the contribution in TEC anomalies due to EQs is more prominent in past reports [4,8,12].

In this paper, a statistical analysis of seismo ionospheric anomalies related to $M_w > 5$ EQs is performed in three different latitudinal zones to determine the morphological characteristics of the TEC anomalies as a result of EQ intensification. The three zones include the equatorial zone (Zone-B) and the mid-to-low/high latitude zones (Zones-A and C). The enhancement and depletion in TEC due to equatorial and non-equatorial anomaly is correlated with EQs, since EQs trigger enhancements and depletion at different latitudes [12,13,36]. In this paper, we have investigated EQs as a main driving force to anomalous TEC at the equatorial and mid-to-low/high latitudinal regions.

The results have shown that, for all latitudinal regions, seismo

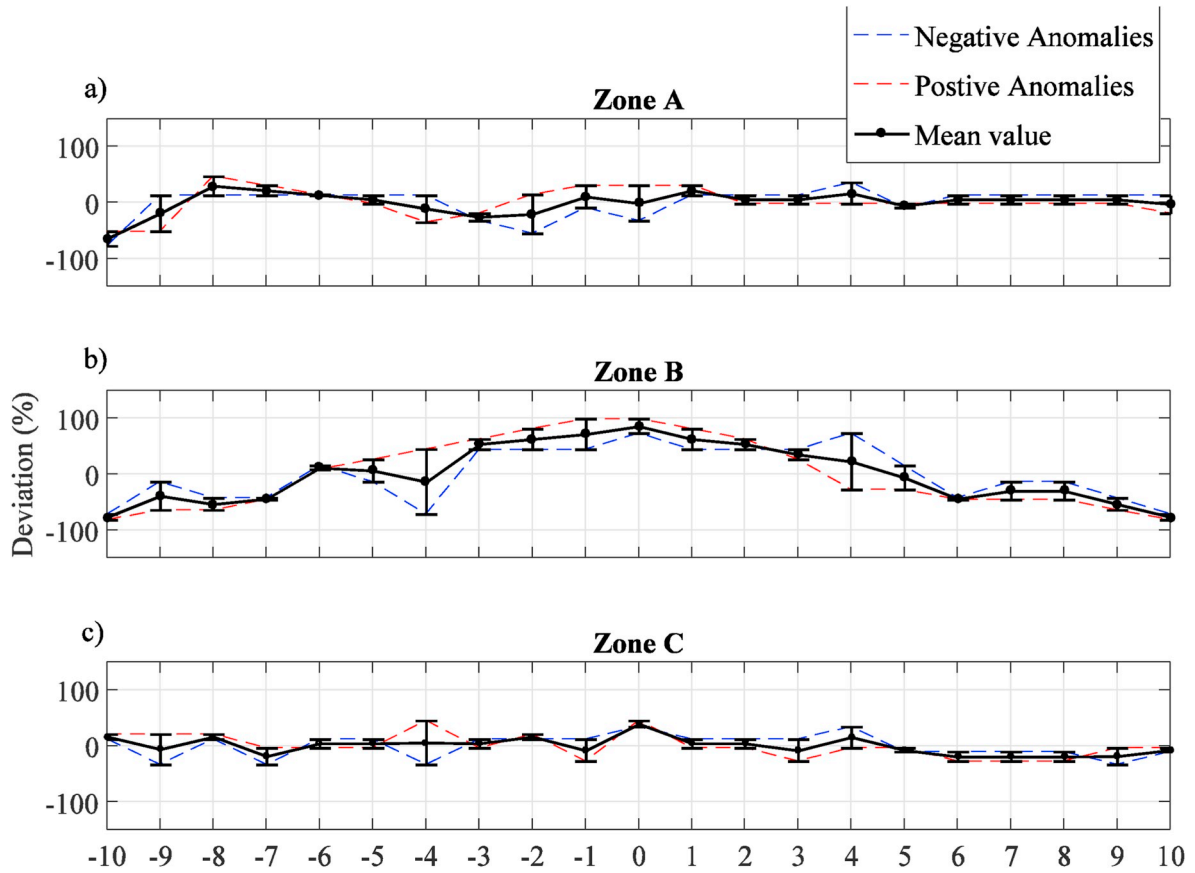


Fig. 8. The deviation percentage of the positive and negative TEC anomalies related to EQs in the three different zones of the world.

ionospheric anomalies are more intense within 5 days before and after the main shock. TEC anomalies of less intensity also occur during 5–10 days interval. However, the strength of these anomalies is smaller than those occurred during 5 days before the main shock (Fig. 5). Furthermore, our study has shown more intense TEC anomalies before $M_w > 6.0$ in Zone-B, probably related to higher TEC along the equatorial regions. Similarly, EQs of $M_w > 7.0$ caused TEC anomalies in Zones-A and C probably due to a weaker TEC far from the equatorial

crest regions. In this study, we reveal that TEC anomalies are associated with EQs of focal depth ≤ 220 km in Zone-B. On the other side, TEC anomalies in Zones-A and C are only limited to EQs at a focal depth less than 120 km. This suggests that the focal depth of the EQ epicenter plays an important role in the occurrence of the seismo ionospheric anomalies. The results shown in this paper suggests that shallow EQs are contributing more energy to the atmosphere, creating ionospheric anomalies, and then disturbing the path of electromagnetic signals.

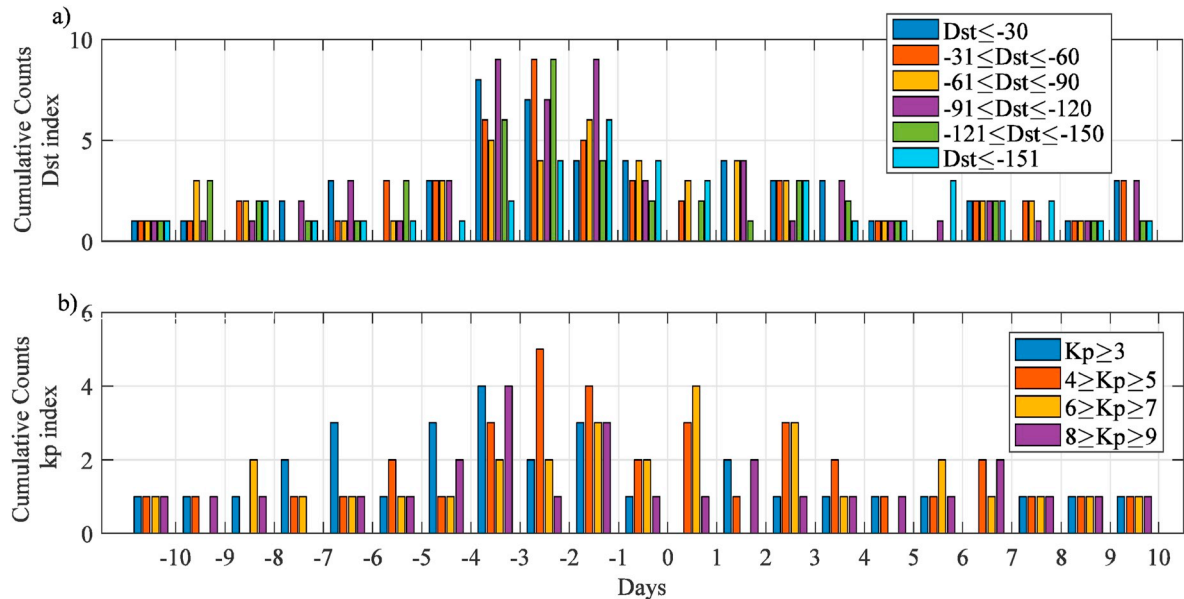


Fig. 9. TEC anomalies related to different geomagnetic storms indices of different intensities.

Previous reports on seismo ionospheric anomalies in the equatorial regions have reported ionospheric anomalies from TEC and other satellite measurements [13,15]. These authors showed the evidence for seismo ionospheric anomalies before and after the EQs in the equatorial and mid-latitudinal sectors, but no comprehensive statistical analysis of large data set for different latitudinal regions were presented as we have shown in this paper.

The goal of this study is to find the time interval for monitoring TEC anomalies before and after the main shocks of EQs. As discussed above, we have seen sharp perturbations within 5 days before and after the EQ events, and severe TEC enhancements lasted even after the 5th day. The evidence of seismo ionospheric anomalies within 5 days before and after the main shock agrees with the findings of Liu et al. [12] and Kon et al. [15]. These authors pointed out an abrupt TEC variations during 5 days before the main shock, lasting few days after main shock.

4. Conclusions

In this paper, TEC intensification and depletion in the form of seismo ionospheric anomalies prior to EQs have been studied for 1182 events globally during 1998–2019 for three different latitudinal zones. We exclude variations caused by geomagnetic storms and that is estimated by a separate analysis. The results have shown evident existence of TEC anomalies within 5 days before and after the EQs, being the pre-seismic ionospheric anomalies stronger than the post-seismic ionospheric anomalies. Furthermore, the EQs of $M_w > 6.0$ and focal depth < 220 km caused TEC anomalies in the equatorial to low latitude regions (Zone-B), whereas the EQs having $M_w > 7.0$ and focal depth < 120 km can produce TEC anomalies in the North and South mid to high latitudes (Zones-A and C). Moreover, we have found that the percentage of occurrence of positive anomalies is larger than the negative anomalies. Geomagnetic storms of high intensity caused more variations than low intensity storms, as evident from TEC anomalies. This study provides a deep insight into different morphological characteristics of TEC anomalies due to EQs; however, more work is required to investigate the phenomena of EQ forecasting.

Declaration of competing interest

The authors declare no conflict of interest with any researcher and/or funding agency.

Acknowledgments

The authors are also very grateful to IGS for the TEC data, and OMNI NASA and Kyoto University for providing space weather data. There is no funding received for this manuscript. We thank the Editor and anonymous reviewer for useful comments in improving the paper.

References

- [1] S. Pulintets, A. Legen'ka, Spatial-temporal characteristics of large scale disturbances of electron density observed in the ionospheric f-region before strong earthquakes, *Cosmic Res.* 41 (2003) 221–230, <https://doi.org/10.1023/A:1024046814173>.
- [2] J. Thomas, J. Huard, F. Masci, A statistical study of global ionospheric map total electron content changes prior to occurrences of $M \geq 6.0$ earthquakes during 2000–2014, *J. Geophys. Res.: Space Phys.* 122 (2017) 2151–2161, <https://doi.org/10.1002/2016JA023652>.
- [3] M. Shah, A. Calabia, M.A. Tariq, J. Ahmed, A. Ahmed, Possible ionosphere and atmosphere precursory analysis related to $M_w > 6.0$ earthquakes in Japan, *Remote Sens. Environ.* 239 (2020) 111620, <https://doi.org/10.1016/j.rse.2019.111620>.
- [4] M. Shah, S. Jin, Statistical characteristics of seismo-ionospheric GPS TEC disturbances prior to global $M_w \geq 5.0$ earthquakes (1998–2014), *J. Geodyn.* 92 (2015) 42–49, <https://doi.org/10.1016/j.jog.2015.10.002>.
- [5] F. Ke, Y. Wang, X. Wang, H. Qian, C. Shi, Statistical analysis of seismo-ionospheric anomalies related to $M_s > 5.0$ earthquakes in China by GPS TEC, *J. Seismol.* 20 (2016) 137–149, <https://doi.org/10.1007/s10950-015-9516-x>.
- [6] F. Zhu, F. Su, J. Lin, Statistical analysis of TEC anomalies prior to $M_6.0+$ earthquakes during 2003–2014, *Pure Appl. Geophys.* (2018) 1–10, <https://doi.org/10.1007/s00024-018-1869-y>.
- [7] S. Pulintets, K. Boyarchuk, *Ionospheric Precursors of Earthquakes*, Springer, 2004.
- [8] J. Liu, Y. Chen, Y. Chuo, H. Tsai, Variations of ionospheric total electron content during the Chi-Chi earthquake, *Geophys. Res. Lett.* 28 (2001) 1383–1386, <https://doi.org/10.1029/2000GL012511>.
- [9] J. Ahmed, M. Shah, W.A. Zafar, M.A. Amin, T. Iqbal, Seismoionospheric anomalies associated with earthquakes from the analysis of the ionosonde data, *J. Atmos. Sol. Terr. Phys.* 179 (2018) 450–458, <https://doi.org/10.1016/j.jastp.2018.10.004>.
- [10] M.A. Tariq, M. Shah, M. Hernández-Pajares, T. Iqbal, Pre-earthquake ionospheric anomalies before three major earthquakes by GPS-TEC and GIM-TEC data during 2015–2017, *Adv. Space Res.* (2019), <https://doi.org/10.1016/j.asr.2018.12.028>.
- [11] C. Kuo, L. Lee, J. Huba, An improved coupling model for the lithosphere-atmosphere-ionosphere system, *J. Geophys. Res.: Space Phys.* 119 (2014) 3189–3205, <https://doi.org/10.1002/2013JA019392>.
- [12] J.-Y. Liu, Y. Chen, Y. Chuo, C.-S. Chen, A statistical investigation of preearthquake ionospheric anomaly, *J. Geophys. Res.: Space Phys.* 111 (2006), <https://doi.org/10.1029/2005JA011333>.
- [13] M. Shah, M.A. Tariq, J. Ahmad, N.A. Naqvi, S. Jin, Seismo ionospheric anomalies before the 2007 M7.7 Chile earthquake from GPS TEC and DEMETER, *J. Geodyn.* 127 (2019) 42–51, <https://doi.org/10.1016/j.jog.2019.05.004>.
- [14] H. Le, J.-Y. Liu, L. Liu, A statistical analysis of ionospheric anomalies before 736 $M_6.0+$ earthquakes during 2002–2010, *J. Geophys. Res.: Space Phys.* 116 (2011), <https://doi.org/10.1029/2010JA015781>.
- [15] S. Kon, M. Nishihashi, K. Hattori, Ionospheric anomalies possibly associated with $M \geq 6.0$ earthquakes in the Japan area during 1998–2010: case studies and statistical study, *J. Asian Earth Sci.* 41 (2011) 410–420, <https://doi.org/10.1016/j.jseas.2010.10.005>.
- [16] S. Saroso, J.-Y. Liu, K. Hattori, C.-H. Chen, Ionospheric GPS TEC anomalies and $M \geq 5.9$ earthquakes in Indonesia during 1993–2002, *Terrestrial, Atmospheric and Oceanic Sci.* 19 (2008).
- [17] Y. Yao, P. Chen, S. Zhang, J. Chen, F. Yan, W. Peng, Analysis of pre-earthquake ionospheric anomalies before the global $M = 7.0+$ earthquakes in 2010, *Nat. Hazards Earth Syst. Sci.* 12 (2012) 575–585, <https://doi.org/10.5194/nhess-12-575-2012>.
- [18] J. Liu, W. Wan, Spatial-temporal distribution of the ionospheric perturbations prior to $M \geq 6.0$ earthquakes in China main land, *Chin. J. Geophys.* 57 (2014) 2181–2189, <https://doi.org/10.6038/cjg20140713>.
- [19] E. Astafyeva, L.M. Rolland, A. Sladen, Strike-slip earthquakes can also be detected in the ionosphere, *Earth Planet Sci. Lett.* 405 (2014) 180–193, <https://doi.org/10.1016/j.epsl.2014.08.024>.
- [20] T. Dautermann, E. Calais, J. Haase, J. Garrison, Investigation of ionospheric electron content variations before earthquakes in southern California, 2003–2004, *J. Geophys. Res.: Solid Earth* 112 (2007), <https://doi.org/10.1029/2006JB004447>.
- [21] E.L. Afraimovich, E.I. Astafyeva, TEC anomalies—local TEC changes prior to earthquakes or TEC response to solar and geomagnetic activity changes? *Earth Planets Space* 60 (2008) 961–966, <https://doi.org/10.1186/BF03352851>.
- [22] E. Thomas, J. Baker, J. Ruohoniemi, A. Coster, S.R. Zhang, The geomagnetic storm time response of GPS total electron content in the North American sector, *J. Geophys. Res.: Space Phys.* 121 (2016) 1744–1759, <https://doi.org/10.1002/2015JA022182>.
- [23] A.J. Michael, Random variability explains apparent global clustering of large earthquakes, *Geophys. Res. Lett.* 38 (2011) L21301, <https://doi.org/10.1029/2011GL049443>.
- [24] J. Feltens, The international GPS service (IGS) ionosphere working group, *Adv. Space Res.* 31 (2003) 635–644.
- [25] M. Hernández-Pajares, J. Juan, J. Sanz, R. Orus, A. Garcia-Rigo, J. Feltens, A. Komjathy, S. Schaer, A. Krankowski, The IGS VTEC maps: a reliable source of ionospheric information since 1998, *J. Geodes.* 83 (2009) 263–275.
- [26] D. Roma-Dollase, M. Hernández-Pajares, A. Krankowski, K. Kotlák, R. Ghoddousi-Fard, Y. Yuan, Z. Li, H. Zhang, C. Shi, C. Wang, Consistency of seven different GNSS global ionospheric mapping techniques during one solar cycle, *J. Geodes.* 92 (2018) 691–706.
- [27] M. Shah, S. Jin, Pre-seismic ionospheric anomalies of the 2013 $M_w = 7.7$ Pakistan earthquake from GPS and COSMIC observations, *Geodesy Geodynam.* 9 (2018) 378–387, <https://doi.org/10.1016/j.geog.2017.11.008>.
- [28] I. Dobrovolsky, S. Zubkov, V. Miachkin, Estimation of the size of earthquake preparation zones, *Pure Appl. Geophys.* 117 (1979) 1025–1044, <https://doi.org/10.1007/BF00876083>.
- [29] E. Afraimovich, A. Altynsev, E. Kosogorov, N. Larina, L. Leonovich, Ionospheric effects of the solar flares of September 23, 1998 and July 29, 1999 as deduced from global GPS network data, *J. Atmos. Sol. Terr. Phys.* 63 (2001) 1841–1849, [https://doi.org/10.1016/S1364-6826\(01\)00060-8](https://doi.org/10.1016/S1364-6826(01)00060-8).
- [30] D. Altadill, E. Apostolov, Time and scale size of planetary wave signatures in the ionospheric F region: role of the geomagnetic activity and mesosphere/lower thermosphere winds, *J. Geophys. Res.: Space Phys.* 108 (2003), <https://doi.org/10.1029/2003JA010015>.
- [31] B. Xiong, W. Wan, B. Ning, F. Ding, L. Hu, Y. Yu, A statistic study of ionospheric solar flare activity indicator, *Space Weather* 12 (2014) 29–40, <https://doi.org/10.1002/2013SW001000>.
- [32] W. Gonzalez, J. Joselyn, Y. Kamide, H. Kroehl, G. Rostoker, B. Tsurutani, V. Vasyliunas, What is a geomagnetic storm? *J. Geophys. Res.: Space Phys.* 99 (1994) 5771–5792, <https://doi.org/10.1029/93JA02867>.
- [33] L. Liu, W. Wan, M.-L. Zhang, B. Zhao, B. Ning, Prestorm enhancements in NmF2 and total electron content at low latitudes, *J. Geophys. Res.* 113 (2008) A02311, <https://doi.org/10.1029/2007JA012832>.
- [34] M. Shah, S. Inyurt, M. Ehsan, A. Ahmed, M. Shakir, S. Ullah, M.S. Iqbal, Seismo

- ionospheric anomalies in Turkey associated with $M_w \geq 6.0$ earthquakes detected by GPS TEC and GIM TEC, *Adv. Space Res.* (2020), <https://doi.org/10.1016/j.asr.2020.03.005>.
- [35] J. Ahmed, M. Shah, W.A. Zafar, M.A. Amin, T. Iqbal, Seismoionospheric anomalies associated with earthquakes from the analysis of the ionosonde data, *J. Atmos. Sol. Terr. Phys.* 179 (2018) 450–458, <https://doi.org/10.1016/j.jastp.2018.10.004>.
- [36] M.A. Tariq, M. Shah, M. Hernández-Pajares, T. Iqbal, Pre-earthquake ionospheric anomalies before three major earthquakes by GPS-TEC and GIM TEC data during 2015–2017, *Adv. Space Res.* 63 (2019) 2088–2099, <https://doi.org/10.1016/j.asr.2018.12.028>.
- [37] Z. Li, Y. Yuan, L. Fan, X. Huo, H. Hsu, Determination of the differential code bias for current bds satellites, *IEEE Trans. Geosci. Rem. Sens.* 52 (7) (2014) 3968–3979.
- [38] Z. Li, N. Wang, L. Wang, et al., Regional ionospheric TEC modeling based on a two-layer spherical harmonic approximation for real-time single-frequency PPP, *J. Geodes.* (2019), <https://doi.org/10.1007/s00190-019-01275-5>.